

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Simulate linear convolution and circular convolution on discrete time signals.	
Experiment No: 2	Date: 25-07-2026	Enrollment No: 92400133120

Theory:

Linear convolution and circular convolution are mathematical operations used to combine two signals to obtain a third signal. They are widely used in various applications, such as signal processing, image processing, and audio processing.

Linear convolution calculates the sum of element-wise products of two signals, considering the full range of valid indices. It is typically used for finite-length signals and can produce an output signal that is longer than the input signals.

Circular convolution, on the other hand, calculates the sum of element-wise products of two signals, considering a periodic extension of the input signals. It is commonly used for periodic or infinite-length signals and produces an output signal with the same length as the input signals.

Flowchart:

The flowchart for the program will consist of the following steps:

Define two discrete-time signals (signal 1 and signal 2).

Compute the linear convolution of the two signals using `numpy.convolve`.

Compute the circular convolution of the two signals using `numpy.fft.ifft`.

Plot and display the linear and circular convolution results.

Save the linear and circular convolution results (optional).

Now, let's see the Python program that simulates linear convolution and circular convolution on discrete-time signal

Explanation:

We import the required modules, including `numpy` for array operations and `matplotlib.pyplot` for plotting.

The `linear_convolution` function computes the linear convolution between two signals using `numpy.convolve` with the mode set to 'full'.

The `circular_convolution` function computes the circular convolution between two signals using `numpy.fft.fft` and `numpy.fft.ifft`. It first calculates the FFT of both signals with a specified length and then computes the inverse FFT of the element-wise product.

In the main part of the program, we define two discrete-time signals (`signal1` and `signal2`).

We compute the linear convolution by calling the `linear_convolution` function with `signal1` and `signal2`.

We compute the circular convolution by calling the `circular_convolution` function with `signal1` and `signal2`.

We plot and display the linear and circular convolution results using `matplotlib.pyplot.stem`.

The linear and circular convolution results can be optionally saved to files using `numpy.savetxt`.

You can modify the `signal1` and `signal2` variables to use different discrete-time signals for convolution.

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Simulate linear convolution and circular convolution on discrete time signals.	
Experiment No: 2	Date: 25-07-2026	Enrollment No: 92400133120

Uncomment the np.savetxt lines if you want to save the linear and circular convolution results to text files.

Make sure you have numpy and matplotlib installed (pip install numpy matplotlib) before running this program.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

def linear_convolution(signal1, signal2):
    lin_conv = np.convolve(signal1, signal2, mode="full")
    return lin_conv

def circular_convolution(signal1, signal2):
    if (len(signal1) > len(signal2)):
        sig_length = len(signal1)
    else:
        sig_length = len(signal2)
    fft_signal1 = np.fft.fft(signal1, sig_length)
    fft_signal2 = np.fft.fft(signal2, sig_length)
    circu_convo = np.fft.ifft(fft_signal1 * fft_signal2)
    return circu_convo

signal1 = [1, 2, 3, 4, 5]
signal2 = [2, 4, 6, 8, 10]
linear = linear_convolution(signal1, signal2)
circular = circular_convolution(signal1, signal2)
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(linear)
plt.title("Linear Convolution")
plt.xlabel("Sample")
plt.ylabel("Amplitude")
plt.subplot(2, 1, 2)
plt.stem(circular)
plt.title("circular Convolution")
plt.xlabel("Sample")
plt.ylabel("Amplitude")
plt.tight_layout()
plt.show()
```

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Simulate linear convolution and circular convolution on discrete time signals.	
Experiment No: 2	Date: 25-07-2026	Enrollment No: 92400133120

Expected Output:

A plot will be displayed showing the linear and circular convolution results.

The linear and circular convolution results will be saved to text files (optional).

Feel free to experiment with different discrete-time signals and observe the linear and circular convolution outputs.

